

Depreciation



- 1 Ray purchased a \$9300 computer, which depreciates at a compounded rate of 25% p.a. Answer the following questions without using the depreciation formula.
 - a What is the salvage value of the computer after the first year?
 - b By how much did the computer depreciate in the first year?
 - c What is the salvage value after the second year?
 - d By how much does computer depreciate over the second year only?
 - e What is the total depreciation over the 2 years?
- 2 Neil purchased a \$1300 motorbike which depreciates at a compounded rate of 10% p.a.
 - a Find the total depreciation over the two years.
 - b Find the percentage of the original value remaining after two years. Round your answer to the nearest percentage.
- 3 A car was originally purchased for \$3000, and depreciates at 14% p.a. Use the depreciation formula to calculate its expected value after 8 years.
- 4 A dishwasher selling for \$800, depreciates at 11% p.a.
 - a What percentage of the original value will remain after 1 year?
 - b What percentage of the original value will remain after 2 years? Round your answer to two decimal places.
 - c What percentage of the original value will remain after 3 years? Round your answer to two decimal places.
 - d How many full years will it take for the dishwasher to lose half its original value?
 - e How many years will it take for the dishwasher to lose 90% of its original value?
- 5 After one year, the value of a company's machinery had decreased by \$3900 from \$65 000.
 - a At what rate did the machinery depreciate in the first year?
 - b If the value of the machinery continues to decrease at this rate every year, what will it be worth in another 19 years?
- 6 A car depreciated in value from \$20 000 to \$17 530 in 3 years. Find the annual depreciation rate, as a percentage to two decimal places.
- 7 Han's share portfolio of \$83 000 decreased in value by 14% per month for the first 4 months of the Global Financial Crisis and then 3% per month for the 5 months after that. Find the value of his portfolio after 9 months.

- 8 A laptop depreciated by 21% p.a. and was sold at \$700 after 9 years. Find its original price.
- 9 A microwave that costs \$700 depreciates at 10% p.a. After how many full years will the value of the microwave be less than:
- a \$525 b \$350 c \$70
- 10 Joanne purchased a laptop for \$800, which depreciates at 13% p.a. Find its salvage value after 6 years.
- 11 A camera originally purchased for \$3000, depreciates at a rate of 29% p.a. Use the declining-balance depreciation formula to calculate its expected value after 10 years.
- 12 Fred deposited \$9000 into a stock portfolio. This amount decreased by 11% each year for 10 consecutive years. What was the value of the portfolio after 10 years?
- 13 The government wants to decrease its spending on job creation. Currently, it is spending \$11 million and will decrease it by 2.55% p.a. over the next 8 years. Calculate the government's spending in 8 years' time. Round your answer to the nearest dollar.
- 14 The population of some native bees is declining at a rate of 8% per year. If there are 10 000 bees in a hive now, how many will there be in 5 years?
- 15 Eileen purchased a new bed for \$8000, which depreciates at the rate of 15% p.a. Answer the following questions by repeated multiplication.
- a What is the expected value of the bed after 4 years?
- b What is the total depreciation of the bed after 4 years?
- c What is the percentage of the original value that will remain after 4 years? Give your answer as a percentage to two decimal places.
- 16 Marge purchased new office equipment for a total of \$6560.
- a Calculate the value of the equipment after 3 years if she uses straight line depreciation at 7% per year.
- b Calculate the value of the equipment after 3 years if she uses declining-balance depreciation at 7% per year.
- c Assuming declining-balance depreciation, after how many full years will the equipment be worth \$3000?
- 17 A Macbook currently worth \$1823 was purchased 3 years ago for \$2000. Find the annual depreciation rate, R . Write your answer as a percentage to two decimal places.

- 18 The population of an island was 1 610 000  0. Due to rising sea levels leaving more and more parts of the island underwater, the population fell at the rate of 18% p.a. What was the population in 1989? Write your answer to the nearest whole number.
- 19 A radioactive substance decays by 1.86% each year and there are currently 19 300 grams of the substance present. If there were P grams present 10 years ago, find the value of P to one decimal place.
- 20 Mae purchased an iPod for \$100, which depreciates at 18% p.a. What is its resale value after 4 years?
- 21 A car depreciated in value from \$20 000 to \$17 530 in 3 years.
At what rate per annum, R , did it depreciate? Write your answer as a percentage correct to two decimal places.
- 22 A swimming pool is losing water at the rate of 8% per week due to evaporation. The pool currently holds 490 kilolitres of water.
- a How much water will be lost in the next 26 days? Give your answer in kilolitres and correct to two decimal places.
 - b How many weeks will it take for the pool to become only half full?
- 23 Viewer ratings for a show increased by 5% per episode for the first 7 episodes, and then decreased by 6% per episode for the last 6 episodes. If there were 129 000 viewers during the first episode, how many viewers were there during the last episode? Round your answer to the nearest viewer.
- 24 The number of people with access to basic healthcare is decreasing by 2.4% of the previous year's number. If 1 414 000 people currently have access, how many people will have access to basic healthcare in 7 years? Round your answer to the nearest person.
- 25 A lab has \$1600 worth of radioactive polonium stored away, which depreciates at 0.81% p.a. as it decays. What value of polonium does the lab have after 86 years?
- 26 After 1 year, the value of a company's machinery had decreased by \$8946 from \$63 900.
- a At what rate did the machinery depreciate in the first year? Give your answer as a percentage.
 - b If the value of the machinery continues to decrease by the same rate every year, what will it be worth in another 11 years?

27 Amelia purchased a signed photograph for \$500, which depreciates at 4.34% p.a. After 8 years, using single line depreciation, the signed photograph would be worth \$493.52.

- a** Calculate its resale value after 8 years using the declining-balance method.
- b** Which method of depreciation should Amelia hope her signed photograph follows: straight line method or declining-balance method?

28 Sean is looking to purchase a car that he is planning to re-sell in 3 years. A new luxury car is going to cost \$10 000, which depreciates at 20% p.a. A reliable used car costs \$5000 and depreciates at 10% p.a.

- a** Calculate the resale value after 3 years of the:
 - i** Luxury car.
 - ii** Used car.
- b** Calculate the loss in value after 3 years of the:
 - i** Luxury car.
 - ii** Used car.
- c** Which car has the higher resale value?
- d** Which car has the smallest decrease in value?
- e** Which car should Sean purchase to minimise his losses?

29 A computer depreciated by 16% p.a. and was valued at \$517 after 6 years. The owner of the computer wants to calculate the original cost.

- a** To find the initial value, V_0 , the following equation needs to be rearranged:

$$S = V_0 (1 - r)^n$$

Write the formula for V_0 .

- b** What was the computer's original price?

30 A property depreciated by 4.1% p.a. and was valued at \$901 500 after 29 years. What was its original price?

31 James is purchasing scuba diving gear that costs \$600 and depreciates at 14% p.a. After how many full years will the value of the scuba diving gear be less than:

- a** \$450?
- b** \$300?
- c** \$60?

32 A ball is dropped from a height of 21 m. After each bounce, it reaches half the previous height. Find the height reached after the 5th bounce correct to two decimal places.

- 
- 33** A car salesman received a total commission of \$3496 at the beginning of the month. He expects that since fewer cars are being purchased, his commission will decrease by 12% each month over the next few months. According to his prediction, what will be his commission in 7 months' time?
- 34** Cinema ticket sales for a new movie increased by 3% per day for the first 5 days after its release date, and then decreased by 4% per day for the next 3 days. If \$259 000 in ticket sales were made on the release date, how much was made in sales on the 8th day after its release to the nearest dollar?